GD, SGD and Polyak stepsizes for linear regression on the California housing data

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Idea

The idea is to take some real data, train a model on 80% of it and then see what the result is on the remaining 20%. If the model is a linear regression, the problem is easy to solve exactly, without any gradient descent. This is exactly why the first experiments are done with it: since the exact solution is known, every method can be compared with it.

Data and model

We take the California house prices data, build a linear model and minimize the quadratic loss function

$$f(w) = \frac{1}{n}\sum_{i=1}^{n} f_i(w), \qquad f_i(w) = \frac{1}{2}\big(\langle x_i, w\rangle - y_i\big)^2$$

with different gradient methods. For this problem we know all the constants of the theory, L, $L_{\max}$, $\bar{L}$ and μ, as well as the solution $w^\star$ itself.

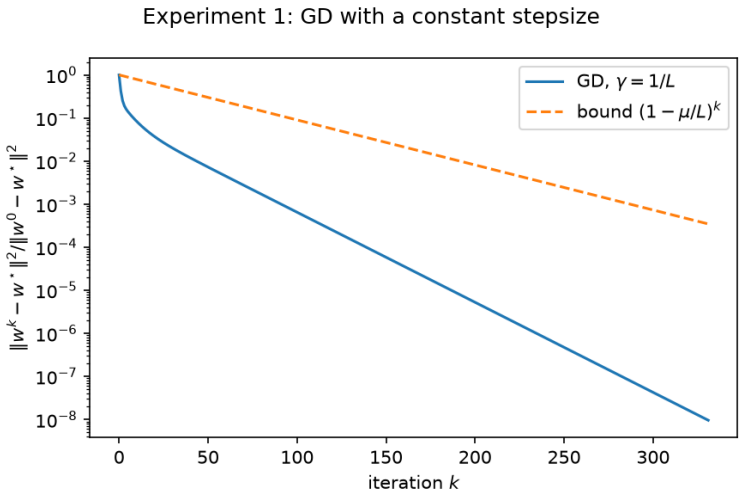
One more note: we take the logarithm of some of the features and clip the outliers to the region $[-3, 3]$. This makes $L_{\max}$ reasonably small, so that it is representative as a theoretical maximum.

The code is the file

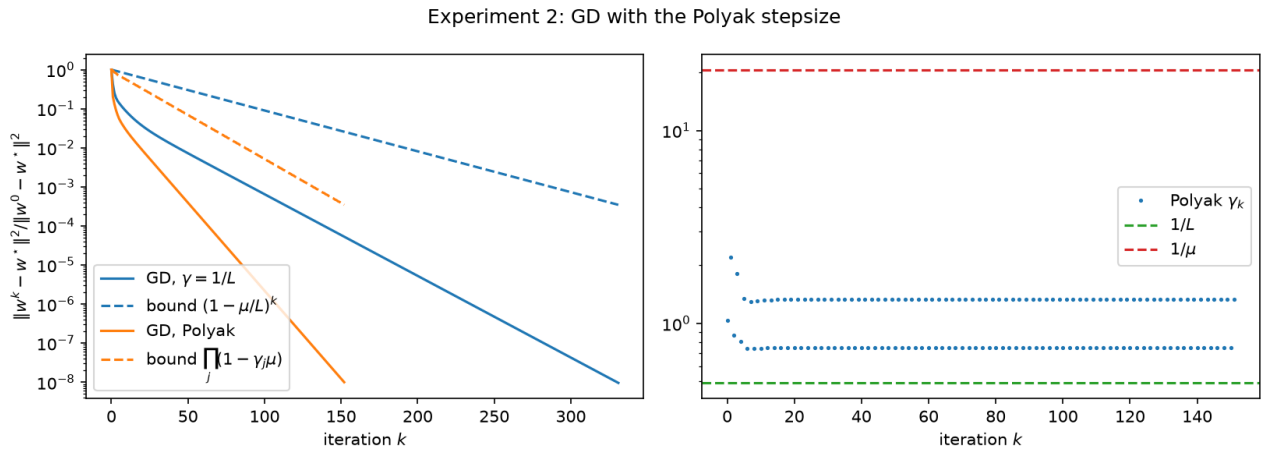
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*Experiments

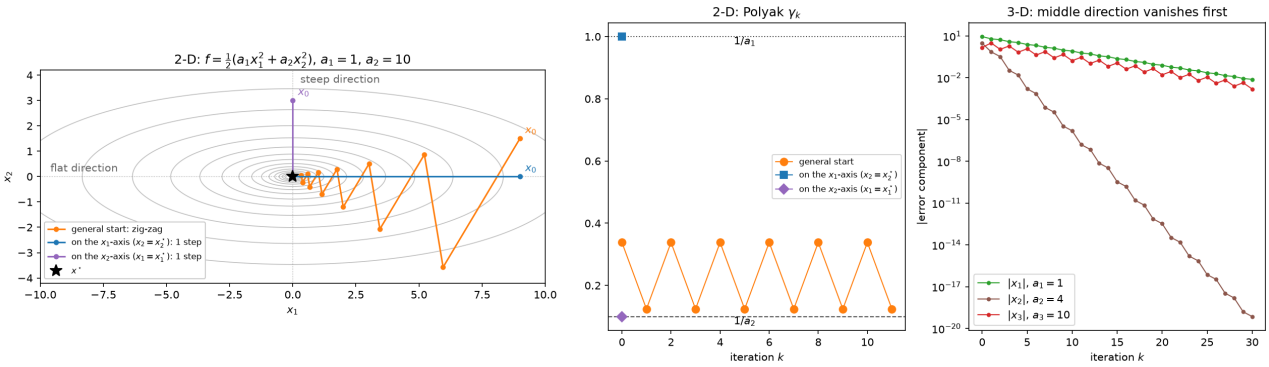
The first experiment is GD with the constant stepsize $\frac{1}{L}$.



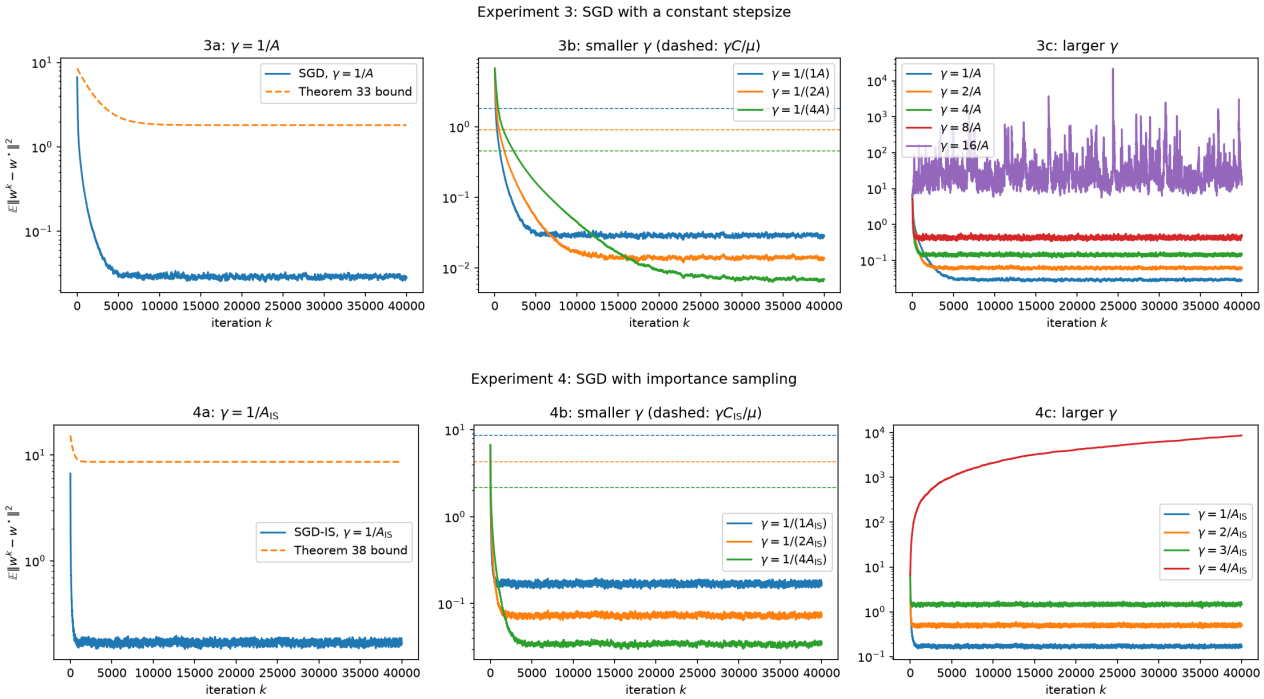
The second one is GD with the Polyak stepsize. The plot shows the theoretical bound and the actual convergence, together with the comparison with the simple GD.



The Polyak stepsizes on the right plot jump between two values. The toy example $f(x) = \frac{1}{2}(a_1 x_1^2 + a_2 x_2^2)$ with $a_1 = 1$, $a_2 = 10$ shows why. One γ_k can not be good for both directions at once: a large step helps in the flat direction x_1 but overshoots in the steep direction x_2, and the next, smaller step corrects it. So the path zig-zags and γ_k ping-pongs. If one coordinate is already correct, e.g. $x_0 = (9, 0)$ or $x_0 = (0, 3)$, then $\gamma_0 = \frac{1}{a_1}$ or $\frac{1}{a_2}$ and the minimum is reached in one step. In more dimensions, e.g. with $a_1 = 1$, $a_2 = 4$, $a_3 = 10$ (last plot), the middle direction vanishes first, and again only the flattest and the steepest directions remain.



After them come SGD-US and SGD-IS with their theoretical maximal stepsizes $\frac{1}{A}$, and also some experiments with other stepsizes. They show how the stepsize changes the convergence speed and the size of the neighbourhood, and what happens when we take a much larger γ than the theory allows.



Comparison

The last experiment is the comparison of all the methods, both by the number of iterations and by the time.

